PERFORMANCE TEST ON Kaplan Turbine

AIM:-

To study the performance of a Kaplan Turbine and plot the following graphs.

1. Efficiency v/s Discharge
2. Head v/s Discharge
3. Input power v/s Discharge
4. Efficiency v/s Brake Power (at ......Kg load)

APPARATUS USED

1. Kaplan Turbine
2. Centrifugal pump to supply water to the turbine.
3. Flow measuring units- venturimeter & manometer.
4. Piping system

PRINCIPLE

Kaplan Turbine : efficiency $\eta = \dfrac{Brake\ Power}{Input\ Power} \times 100\ \%$

Input Power, IP $= \rho\, g\, Q\, H$ watt.

Where,

ρ = Density of water in kg/m^3

g = Acceleration due to gravity in m/s^2

Q = Discharge of water in m^3/s

H = Delivery pressure + Draft tube pressure + Datum head, in m

Discharge, $Q = C_d \times K \times \sqrt{2gh}\ \dfrac{m^3}{s}$

Where,

C_d of Venturimeter = 0.93

Constant, K $= \dfrac{a_1 a_2}{\sqrt{a_1^2 - a_2^2)}}$

a_1 = area of inlet of venturimeter in m^2

Diameter of inlet of venturimeter= 100 mm,

a_2= area of throat of venturimeter in m^2

Diameter of throat of venturimeter = 59.16 mm

Acceleration due to gravity, g = 9.81 m/s^2

h = $(h_2 - h_1) \times 12.6$ (in m of water)

h_1 = Left limb manometer reading in m

h_2 = Right limb manometer reading in m

OBSERVATIONS

Sl. No	Delivery Pressure kg/cm^2	Delivery Pressure m	Draft Pressure mm of Hg	Draft Pressure m	h_1 cm	h_2 cm	$h = (h_1 \sim h_2)$ m	W kg	S kg	w-s kg	Speed rpm	Total Head m	Q m^3/s	BP kW	IP kW	Efficiency %
1																
2																
3																
4																
5	6															
6																

Brake power. $$BP = \frac{2\pi N (W-S) R g}{60000} \text{ kW}$$

Where,

N = Speed of the Turbine in rpm

(W-S) = Effective load in Kg

R = Radius of the brake drum + Radius of the rope

g = Acceleration due to gravity m s^2

Diameter of the brake drum = 300 mm

Diameter of the rope = ...mm

PRECAUTIONS

1. Check whether all the joints are leak proof and water tight.
2. Check the gauge and water seal assembly of the measuring tank and make sure that it is fixed properly.

PROCEDURE

1. Start the pump and keep it at a constant speed of turbine at 1500 rpm. This can be adjusted with the help of the delivery valve provided.
2. The pressure gauge and draft tube vacuum pressure reading has to be noted to find out the total head on the pump
3. Speed of the turbine is noted from the control panel.
4. Discharge of water measured by venturimeter. Note the mercury column manometer readings (h_1 & h_2).
5. The brake power from the turbine is calculated from the reading of brake load (W-S)
6. The above procedure is repeated by gradually increasing the load at a speed of 1500 rpm. (take minimum 6 sets of readings)

RESULT:-

Performance of Kaplan Turbine is found out and the following graphs are drawn.

GRAPHS

1. Efficiency v/s Discharge
2. Head v/s Discharge
3. Input power v/s Discharge
4. Efficiency v/s Brake Power

INFERENCE